Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/30/2008
PI Name:	van der Woerd, Mark Ph.D.		
Project Title:	The Master Switch for Bone Formation: Structural Studies of RunX2		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
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No. of Post Docs:	2	**No. of PhD Degrees:**	0
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received (in October 2007) NCE to 9/30/2008 per A. Chu/ARC (8/08)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Lian, Jane (University of Massachusetts Medical School) Snell, Edward (Hauptman-Woodward Medical Research Institute) Ferree, Darren (BAE SYSTEMS Analytical Solutions) Karr, Laura (NASA/MSFC) Stein, Gary (University of Massachusetts Medical School) Montecino, Martin (University of Massachusetts Medical School)		
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Task Description:	Runx2, which is also called the Core Binding Factor a1 (Cbfa1), is the pivotal molecule for bone formation – the master switch. It regulates osteoblast growth and differentiation at the level of gene transcription, performing these regulatory functions by activation and repression of genes important for bone formation in the osteoblasts. While Runx2 binds to the gene (DNA) with a domain near the N-terminus, the C-terminal region is essential for biological activity. The functional activity of the C-terminus is influenced by a nuclear matrix targeting signal that localizes Runx2 to subnuclear foci for the assembly of transcription factor complexes with a host of interacting co-regulating proteins. We postulate that a significant conformational change occurs within the C-terminal domain of Runx2 upon specific binding of a co-regulatory protein. We hypothesize that the structural changes will differ, dependent on which factor is bound. We are determining both the structure and function of parts of the native and mutant RunX2 molecules, so we will better understand how molecular structure leads to biological function. One of our specific aims is to resolve by X-ray crystallography the three-dimensional structures of the C-terminal domain of wild type and mutant Runx2 molecules, in the absence and presence of regulatory factors. A second specific aim is to experimentally determine the functional activities of Runx2 wild type and point mutants and to identify the mechanisms by which the interactions between Runx2 and other molecules regulate osteoblast differentiation. Significance for NASA: Bone loss occurs at a rate up to 1.5% per month in astronauts while they reside in a reduced gravity environment. It occurs due to an imbalance between bone formation (osteoblast activity on the cellular level) and bone resorption (osteoclast activity). Research on bone loss in long-term reduced gravity missions conducted on astronauts has shown that bone formation is reduced and resorption is unaltered or increased. Our research will help to explain how the activity of the osteoblasts is regulated at the molecular level. Greater understanding of osteoblast activity will lead to future opportunities to better regulate bone formation and therefore reduce bone loss in space and on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Osteoporosis is a recognized disease that affects millions of people, particularly the elderly. It is responsible for hundreds of thousands of fractures annually in the United States alone. Osteoporosis is caused by an imbalance between bone formation and resorption. Each process may be used to restore the balance, for example by anti-resorptive agents or by stimulation of bone formation. The results from our research will contribute to a better understanding of the processes that occur during bone formation. Increasing bone formation is one of the avenues to restore a healthy balance and avoid osteoporosis.
Task Progress:	This report only covers the biophysical characterization and ultimately structural characterization of Runx2, YAP and the complex between Runx2 and Yap. The cell-biological characterization effort under this grant was completed in 2007 and the results have been reported previously. For the study of Runx2 and YAP it is necessary to produce large amounts (10-100mg of each) high-quality, pure and stable protein. We further explored the expression and purification of the relevant domains of runx2 (C-terminal part) and YAP (WW-domain) by recombinant E. coli work. For the case of Runx2, we expressed the protein as a fusion-protein with NusA to increase solubility, while for YAP we established a GST-fusion protein construct. These proteins need to be stable over time because structural studies require long-duration (weeks to months) experiments. We have developed successful protocols to provide milligram quantities of pure and stable proteins, both for NusA-Runx2 and for GST-YAP. The biological activity of Runx2 and YAP consists of, in this case, the binding of these two proteins to each other. Before further characterization of the proteins, we determined the binding to each other by pull down experiments and found that the constructs expressed in, and purified from E. coli do not bind. Although there may be a trivial reason for this, such as steric interference of the tags attached to the active domains, it is also possible that there is a fundamental biochemical reason why the proteins from E. coli do not bind and we investigated potential causes for the lack of biological activity. The most likely reason for lack of biological activity is that YAP needs to be phosphorylated on a (or more than one) tyrosine. Tyrosine phosphorylation is a necessary condition for binding, as published by Zaidi et al (Embo J. (2004), 23, 790-9), and E. coli is normally thought not capable of providing phosphorylation to proteins. We discovered that YAP is in fact phosphorylated specifically on tyrosine during expression in E. coli, but our methods do not allow us to determine if this protein modification is consistent (on all molecules and on all pertinent tyrosines). Our discovery is consistent with recent insight into the existence and function of native tyrosine kinase in E. coli. There are multiple possible scenarios why the proteins we produced do not bind, one of which is that tyrosine phosphorylation on YAP is necessary but not sufficient for the correct interaction of these proteins. Investigations are underway to determine what keeps Runx2 and YAP from interacting. When appropriate interactions have been established, structural investigation of Runx2 and YAP will continue.
Bibliography Type:	Description: (Last Updated: 08/23/2007)